Sean Yépez · ME344 Summer 2026 · Final Project, Option 2

github.com/seanyepez/me344ricechem

Problem: grading consumes the time meant for teaching

In one measured TA workday, 7 of 8 paid hours went to grading. Treemarks uses AI for a first rubric-verification pass; people review the exceptions.

Sean Yépez · Dr. Mourad Bouache · Google

AI checks every rubric decision → the TA reviews uncertain cases → students receive feedback sooner

RICECHEM · THE SUPERVISED BENCHMARK

Real chemistry-exam answers graded against instructor rubrics: 1,264 responses · 4 questions · 27 rubric items → 8,392 TA-labeled TRUE/FALSE decisions (6,700 train · 831 validate · 861 sealed test). One decision = one answer + one criterion → the TA's label.

- Sonkar et al. (2024) · introduces RiceChem and rubric-based grading of long-form answers
- Rao & Callison-Burch (2026) · Autorubric, rubric-based LLM evaluation
- Ferrer et al. (2026) · when to trust LLM graders

RESEARCH QUESTIONS

RQ1 · Quality: can tuning make a local model accurate enough for rubric grading?

RQ2 · Systems: how do CPU, GPU, and TPU change latency, throughput, and utilization?

RQ3 · Operations: which configuration meets the quality bar in under 24 hours at lowest cost?

PROPOSAL · SYSTEM + EXPERIMENT

Containerize and run a supervised experiment on three compute lanes

ORCHESTRATE

Docker + GKE Job

pinned image · reusable K8s manifest

STORAGE + I/O

GCS + range cache

authorized data · checkpoints · aggregate receipts

TRAIN + SERVE

CPU + A100
PyTorch / QLoRA

CUDA · vLLM dynamic batching

TPU v5e-8
JAX / XLA

compiled train / serve step

EXPERIMENT DESIGN · 8,392 TA-LABELED DECISIONS

TRAIN · 6,700 · 80% update adapters	VALIDATE 831 · 10%	TEST 861 · 10%

Validation selects the checkpoint; the test split stays sealed.

Measurements: instrument the workload—and disclose what is missing

CONTROLLED WORKLOAD

861 held-out rubric decisions
same manifest · prompt · parser · output cap · checkpoint

concurrency 1 + 24

EXECUTION LANES	RUNTIME	CAPTURED SIGNALS	
CPU 16 vCPU baseline	**Transformers** host memory · process telemetry	CPU utilization p50 / p95	peak RSS decisions / s
GPU one A100 40 GB	**PyTorch · CUDA · vLLM** NVML · batching · endpoint timing	GPU utilization load time	peak VRAM tokens + decisions / s
TPU v5e-8 slice	**JAX / XLA · vLLM-TPU** XProf · compile cache · GCS-FUSE	TPU utilization compile time	peak HBM cache + I/O

COMPARISON GATE

Compare hardware only when checkpoint, prompt, parser, output cap, batching, and concurrency match.

WORKLOAD
861 decisions · same prompt/parser/output cap · c=1 and c=24

COMPUTE
wall time · p50/p95 · decisions/s · wall/step time as cycle proxy

MEMORY
CPU mCPU + RSS · A100 NVML + VRAM · TPU host memory only

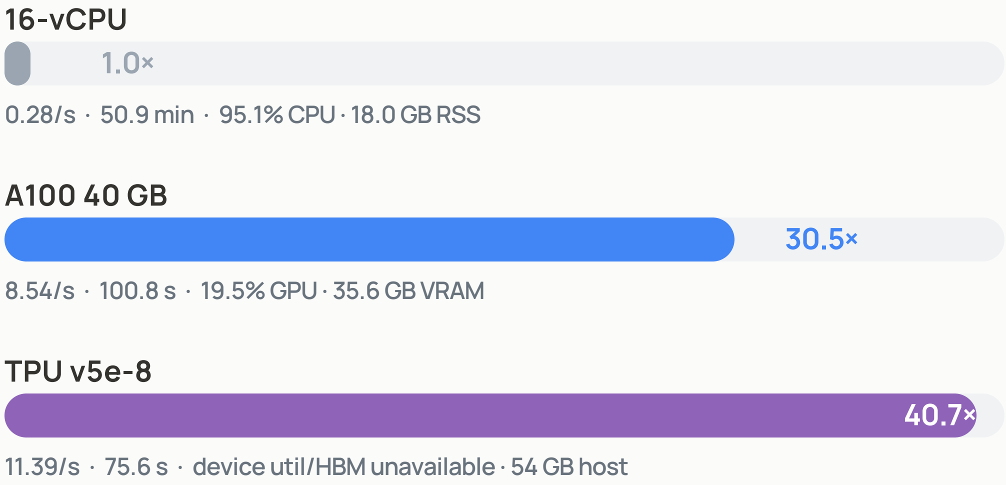
INPUT / I/O
GCS range cache fixed startup; load and XLA compile time not isolated

TPU device utilization + HBM were not enabled in this run. Remedy: add XProf / TensorBoard TPU profiling (or Cloud TPU device metrics), profiler hooks, and service-account permissions; report unavailable values rather than estimate them.

RESULTS · CONTROLLED 4B SERVING PROFILE

Accelerators cut single-stream grading time 30–41×

THROUGHPUT SPEEDUP · CONCURRENCY 1 · CPU = 1×



CONCURRENT SERVING · c = 24
A100 123.9/s
$0.007 per 861 decisions
TPU 98.4/s
$0.023 per 861 decisions
These are one-endpoint measurements. Add GKE runners/endpoints as queue depth grows; cluster throughput is bounded by quota and coordination overhead.

QUALITY GATE · ≈72% AGREEMENT
Gemma 4B failed the practical grading bar.
TA re-review overhead would erase the throughput benefit.

Same 861-decision workload and configured merged-checkpoint path. CPU c=24 did not complete under memory/tunnel pressure. Costs are active-time list-price proxies; TPU device telemetry was unavailable.

CONCLUSION · QUALITY-GATED SYSTEM DESIGN

ME344 moved Treemarks from a local prototype to a repeatable cloud ML experiment

BEFORE	SYSTEM BUILT	MODEL LESSON	OPERATING TARGET
Local prototype	**Docker + GKE + GCS**	**27B shows promise**	**≤24 h**
agent + runner	CPU · A100 · TPU	4B fails quality bar	8 h would be delightful

Meet the grading-quality threshold → choose the smallest system under 24 hours → minimize cost.

BOTTLENECK

27B serving was unbatched; training also exposed accelerator-memory and object-store startup pressure.

NEXT · batch the 27B endpoint; profile 27B on TPU; test leave-one-question-out transfer; compare per-assignment training economics with frontier inference; add runners only when queue depth threatens the 24-hour promise.